1. Attachment

   Attachment is the adherence of a bacterium to the cell membrane of the phagocytic cell. Some bacteria are easily attached to the phagocytic cell. Example , *Mycobacterium tuberculosis*.

2. Phagosome formation

   After attachment the phagocyte extend small pseudopodia around the infecting bacterium. The pseudopodia fuse to form an endosome which contains a bacterium surrounded by the cell membrane. This structure is called as phagosome.

3. Phagolysosome formation

   After engulfment, the lysosome containing the hydrolytic enzymes fuses with phagosome to form phagolysosome. In this step, lysosomal enzymes are discharged to phagosome which is vital for the lysis of bacteria.

4. Lysis

   A number of antimicrobial and cytotoxic substances produced by phagocytes can destroy phagocytosed microorganisms.

Oxygen dependent killing mechanisms

During phagocytosis, a metabolic process known as the 'respiratory burst' occurs in activated phagocytes. This process results in the activation of a membrane bound oxidase that catalyzes the reduction of oxygen to superoxide anion, a reactive oxygen intermediate that is extremely toxic to ingested microorganisms. Activated phagocytes begin to express high levels of nitric oxide synthase (NOS), an enzyme that oxidizes L-arginine to yield L-citrulline and nitric oxide (NO). Much of the antimicrobial activity of phagocytes against bacteria, fungi, parasitic worms, and protozoa is due to nitric oxide and substances derived from it.

Oxygen independent killing mechanisms

Activated phagocytes also synthesize lysozyme and various hydrolytic enzymes whose degradative activities do not require oxygen. In addition, activated phagocytes produce a group of antimicrobial and cytotoxic peptides, commonly known as defensins. Cathepsin G is an example for defensins. These molecules are cysteine-rich cationic peptides that form circularized defensin which inturn form ion-permeable channels in bacterial cell membranes to lyse bacteria. Lactoferrin chelates iron from the medium and prevents the growth and proliferation of iron dependent bacteria. Lysozyme splits mucopeptide in bacterial cell wall and lysed bacteria.

5. Exocytosis

   Finally the killed organisms are digested by hydrolytic enzymes and the degraded products are released to the exterior by the process of exocytosis.

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